

ERIC LIU

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Education

University of Toronto

Sep 2022 – May 2027

Bachelor of Applied Science in Computer Engineering with PEY Co-op

Toronto, Ontario

- On track to graduate with Minors in Artificial Intelligence Engineering and Engineering Business
- Dean's List Scholar

Technical Skills

Programming Languages: Python, C++, C, SQL, Java, JS, Verilog, MATLAB, Assembly, Angular, HTML/CSS
Frameworks & Tools: Stable Baselines3, PostgreSQL, PyTorch, Gym, Git, JSON, Docker, Linux, NumPy, AWS, GCP
Strong skills in advanced DSA: Graph Theory, Dynamic Programming, Monotonic Structures, Greedy Algorithms

Experience

Scotiabank – Quantitative Trading Developer Co-op

May 2025 – April 2026

Electronic Execution Services (Algorithmic Trading) – Global Banking and Markets

Toronto, Ontario

- Tech Environment:** SQL, Python, Angular, Pandas, Bloomberg, TCA, Market Making, Reinforcement Learning.
- Enhanced VWAP execution through reinforcement clustering and GMM modeling across Canadian flows, applying statistical analysis to strengthen quantitative modeling and curve stability.
- Developed the Outage App, a Bloomberg-integrated diagnostic system that identifies and analyzes failed algorithmic executions (e.g., market halts, liquidity limits, client caps) and automates refund calculations using trade specific rates; achieved significant performance by implementing vectorized operations and bitmasking for large scale outage windows.
- Improved Transaction Cost Analysis, Foreign Exchange and Commission reconciliation pipelines, and developed automated client-facing analytics tools, including commission trackers and client billings.
- Collaborated with the Program Trading team on index modeling and rebalancing workflows
- Working on a reinforcement learning driven internal market-making system that intercepts retail order flow before external routing. Improving overall profitability and reducing market impact.

Big Tuna - Software Developer

Aug 2020 – May 2024

Seasonal Game Developer

Oakville, Ontario

- Tech Environment:** JavaScript, JSON, SQL, Developer Portal, Game Design, Git, Software Testing.
- Part of a three-person team that created and maintained a highly engaging Discord bot with more than 23,000 players.
- Integrated SQL database updates corresponding to new content, and conducted comprehensive pre-deployment testing.

Research

Market Making Via Reinforcement Learning Research

May 2023 – Aug 2024

Research Assistant Under Professor Wei Xu

Toronto, Ontario

- Tech Environment:** Python, Stable Baselines (DDPG and DQN algorithms), Linux, Pandas, Gym, PyTorch.
- Researched and developed a profitable market-making agent for cryptocurrency exchange using Deep Deterministic Policy Gradient (DDPG) and Deep Q-Network (DQN), understanding its underlying mathematical and financial constructs.
- Crafted a sophisticated machine learning environment that mimics real-world market conditions and scenarios, integrating over 100 million tradebook data entries, paired with a dynamic reward framework that reacts and gives appropriate response to the model's decisions, facilitating continuous learning and strategic optimization.

Extracurricular

University of Toronto Aerospace Design Team

Jul 2023 – Present

Liquid Propulsion and Aerodynamics Division in Rocketry

Toronto, Ontario

Ontario Engineering Competition (OEC)

January 2024

Represented the University of Toronto for programming competition

Kingston, Ontario

University of Toronto Engineering Kompetition (UTEK)

November 2023

First place in programming competition (Team of 4)

Toronto, Ontario

Projects

Fungi Identification Project using Deep Learning | Python, Git, PyTorch, NumPy, Pandas

Jul 2024

- Developed a comprehensive Convolutional Neural Network (CNN) model that identifies various species of fungi, trained on over 10,000 images, achieving model accuracy of 81% compared to baseline accuracy of 65%.

GIS Mapping System | C++, GTK, Git, Multithreading, Heuristic Algorithms

May 2024

- Implemented map functionalities, graphic rendering, and multithreading optimizations, and utilized time precomputation methods to enhance performance, reducing time complexity.